

MULTI-WINNER VOTING: REPRESENTATION, MANIPULATION, DELIBERATION

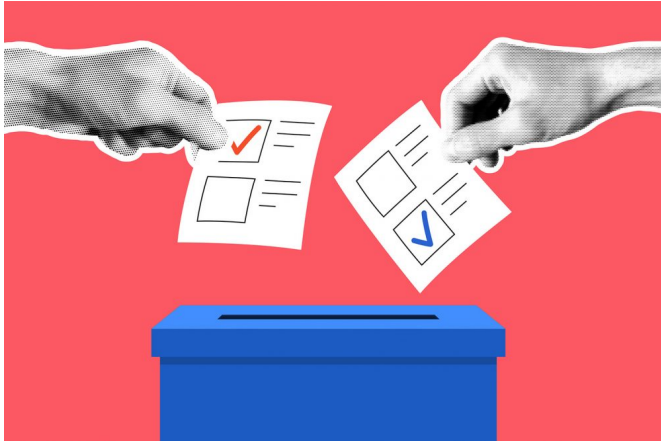
PhD Defense of **Oliviero Nardi**

DBAI, TU Wien

Multi-Winner Voting:
Representation, Manipulation, Deliberation

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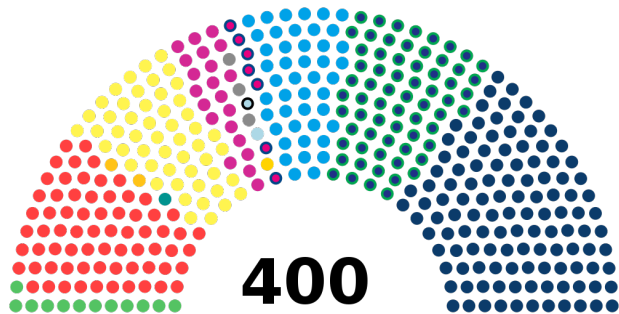
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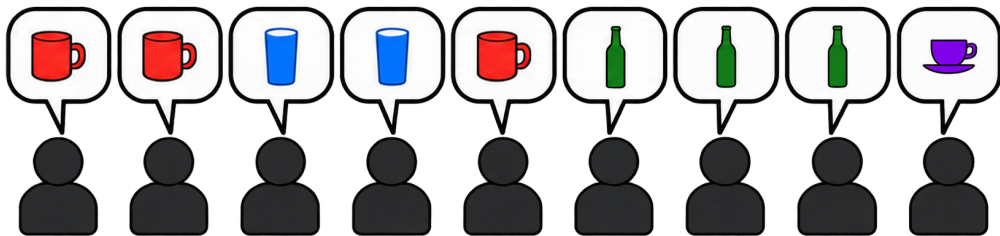
Multi-Winner Voting:

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MajoritarianMango

In systems with PR, opinions are too strongly divided and nothing gets done.

ProportionalParrot

A party's share of seats in the House of Commons should reflect its share of the popular vote.

111 

PastTheP

FPTP results

39 

SingleTransfer

I like the simplicity of stating my favorite candidate, as well as altern

JohnCitizen

PR will reduce wild policy swings and result in more long-lasting policies.

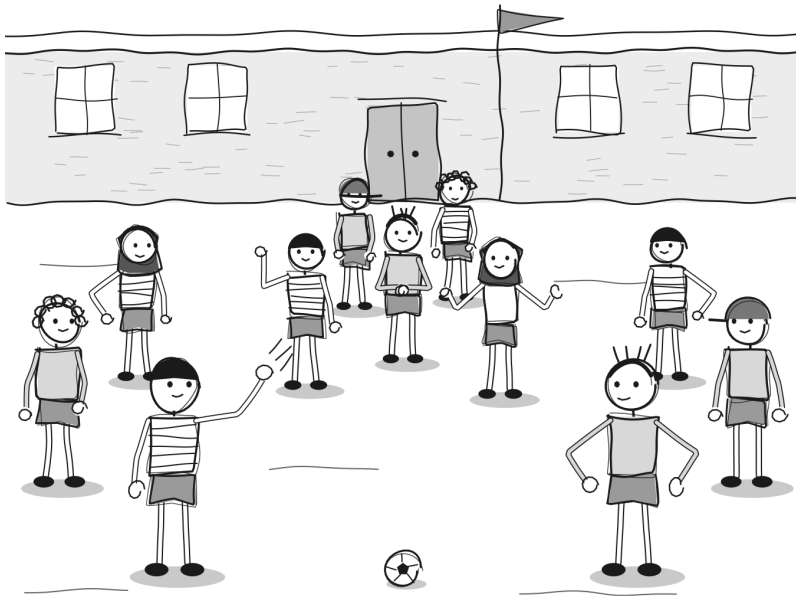
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VDHondt41

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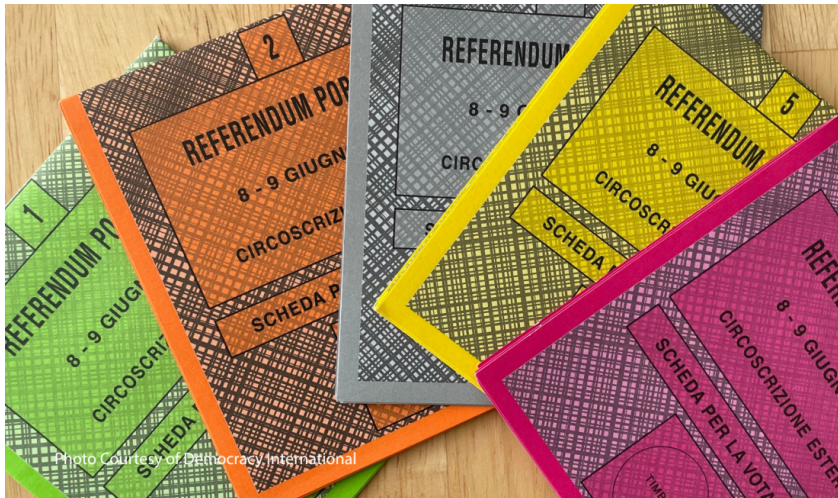


Photo Courtesy of Democracy International



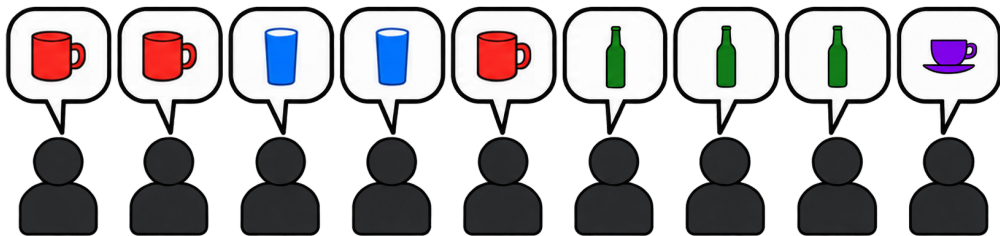
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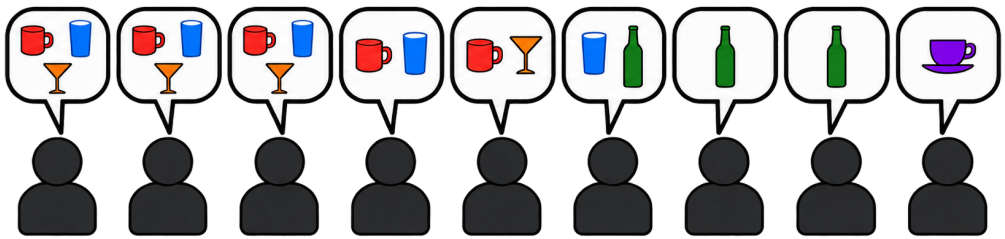
Multi-Winner Voting:

Representation, Manipulation, Deliberation

(Approval-Based)
Multi-Winner Voting:

Representation, Manipulation, Deliberation





**Vote for any number
of options.**

☐

Joe Smith

☒

John Citizen

☐

Jane Doe

☐

Fred Rubble

☒

Mary Hill

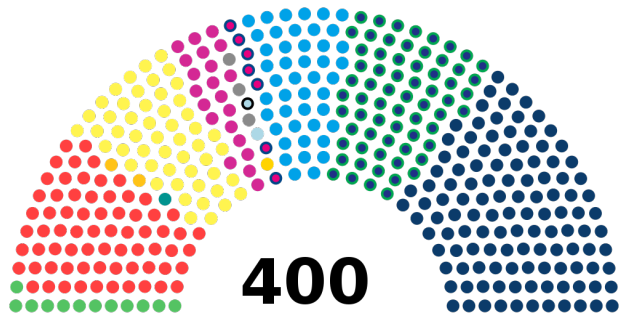
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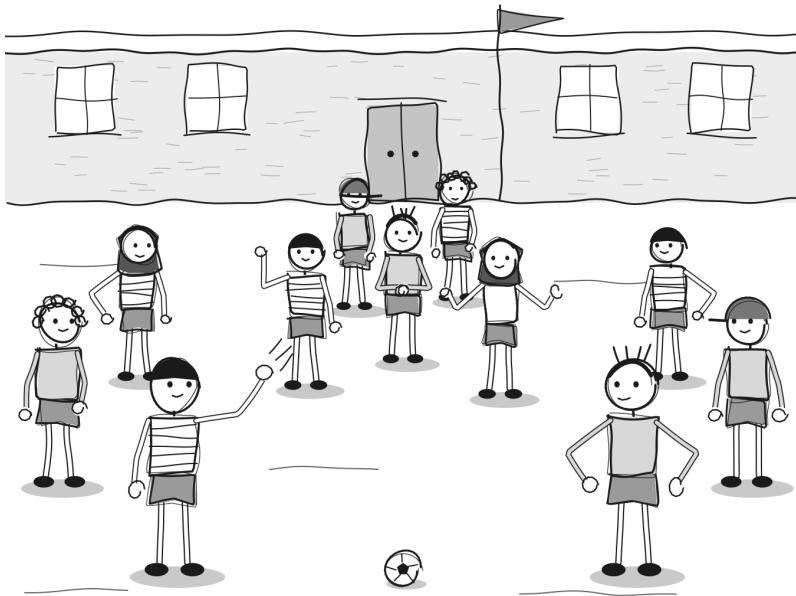
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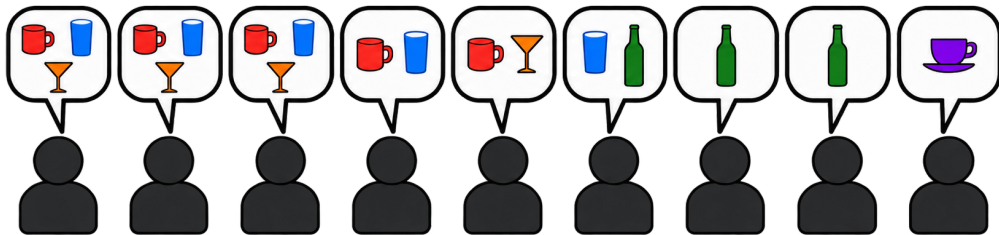
Multi-Winner Voting:

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1. REPRESENTATION



COMMITTEE VOTING, INFORMALLY



COMMITTEE VOTING, FORMALLY

- Voters $N = \{1, 2, \dots, n\}$
- Alternatives $C = \{a, b, \dots\}$ where $|C| = m$
- Target committee size $k \in \mathbb{N}$
- Each voter $i \in N$ submits an approval ballot $A_i \subseteq C$
- Collectively, we have a profile of preferences $\mathbf{A} = (A_1, A_2, \dots, A_n)$
- A voting rule F maps each profile $\mathbf{A}$ to a winning coalition $W \subseteq C$ with $|W| = k$

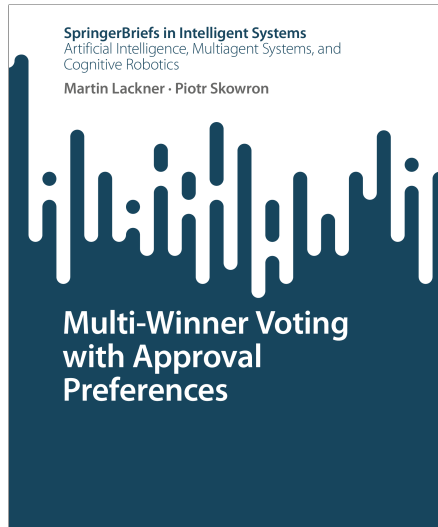
Additionally, we define $N(c) = \{i \in N : c \in A_i\}$.

THE STANDARD CONCEPTION: AVOIDING UNDERREPRESENTATION

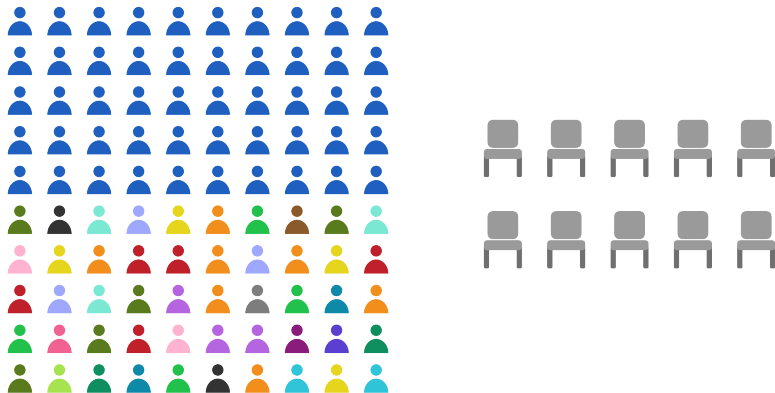
‘if a cohesive group of voters makes up $x\%$ of the population,
then it should get at least $x\%$ of the seats’

AVOIDING UNDERREPRESENTATION: STATE OF THE ART

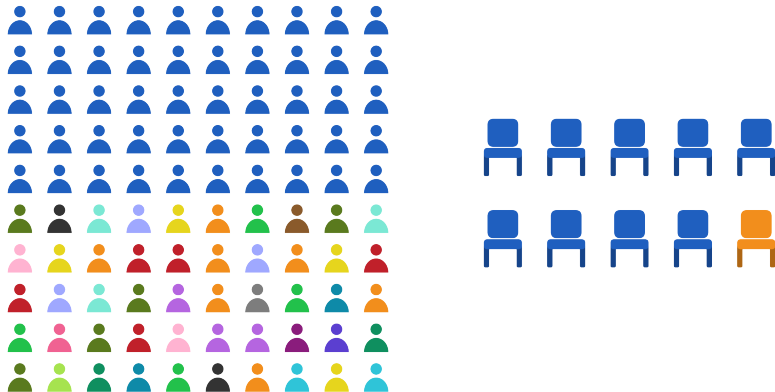
- Well-understood and formalised
 - Extended Justified Representation
 - Priceability
 - ...
- Natural and appealing rules provably satisfy it
 - Proportional Approval Voting
 - Method of Equal Shares
 - ...
- **We argue:** Not the last word on representation!



A PROBLEM WITH ‘REPRESENTATIVE’ RULES



A PROBLEM WITH 'REPRESENTATIVE' RULES



ANOTHER CONCEPTION: AVOIDING OVERREPRESENTATION

‘if a cohesive group of voters makes up $x\%$ of the population,
then it should **not** decide more than $x\%$ of the seats’

WHY AVOIDING OVERREPRESENTATION?

- Fairness:
 - Prevents groups from having ‘too much’ power
 - Avoids ‘envy’
- Might prevent strategic ‘bullet voting’
- Technical applications
 - Safety in blockchain technologies

FORMALISING OVERREPRESENTATION

‘if a cohesive group of voters makes up $x\%$ of the population,
then it should not decide more than $x\%$ of the seats’

OVERREPRESENTATION: TENTATIVE DEFINITION

In a winning coalition W , for every $c \in W$,
the group of voters $N(c)$ should not have

$$|A_i \cap W| > \left\lceil k \frac{|N(c)|}{n} \right\rceil \text{ for all } i \in N(c).$$

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THE DEFINITION HAS A BUG

Say we want to select $k = 6$ candidates. Consider the following profile:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Each voter can get at most $\lceil 6 \cdot \frac{1}{3} \rceil = 2$ candidates. Any committee overrepresents some voter:

- $\{a, b, c, d, e, f\}$ overrepresents voter 1 and voter 2
- $\{a, b, c, d, e, g\}$ overrepresents voter 1
- etc...

JUSTIFIED UPPER QUOTA (JUQ)

In a winning coalition W , for every $c \in W$, if the group of voters $N(c)$ has

$$|A_i \cap W| > \left\lceil k \frac{|N(c)|}{n} \right\rceil \quad \text{for all } i \in N(c),$$

then there cannot be $d \in C \setminus W$ and $N' \subseteq N(d)$ such that

$$|A_i \cap W \setminus \{c\} \cup \{d\}| \leq \left\lceil k \frac{|N'|}{n} \right\rceil \quad \text{for all } i \in N'.$$

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JUQ FIXES THE BUG...

Let $k = 6$ and consider again:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Recall: each voter can get at most $\lceil 6 \cdot \frac{1}{3} \rceil = 2$ candidates.

Committee $\{a, b, c, d, e, g\}$ **satisfies** JUQ:

- **Voter 1** is overrepresented, must kick out c
- But: **voter 2** already at quota
- Thus, no need to swap c with f

...BUT NOT TRIVIALY

Let $k = 6$ and consider again:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Recall: each voter can get at most $\lceil 6 \cdot \frac{1}{3} \rceil = 2$ candidates.

Committee $\{a, b, c, d, e, f\}$ **fails** JUQ:

- **Voter 1** is overrepresented, must kick out c
- **Voter 3** is underrepresented
- Thus, must swap c with g

WHICH RULES SATISFY JUQ?

Consider the standard family of Thiele rules:

Definition

Given a non-increasing $\omega: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$, the ω -Thiele rule picks a committee maximising

$$\sum_{i \in N} \sum_{\ell=1}^{|A_i \cap W|} \omega(\ell)$$

Terrifying news:

Proposition

No Thiele rule satisfies JUQ.

EXTENDING THIELE RULES

We define **composite Thiele rules**, which work in stages of tiebreaking:

Definition

Given a **sequence** of non-increasing $\omega_1, \omega_2, \dots: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$, the corresponding composite Thiele rule picks a committee as follows:

1. First, select **all** committees maximising the score of ω_1 ;
2. Among those, select all committees maximising the score of ω_2 ;
3. Repeat until you are left with a single committee (or without scoring functions).

Theorem

*There is a **unique** composite Thiele rule that satisfies JUQ: we called it **Adams Approval Voting**.*

ADAMS APPROVAL VOTING: DEFINITION

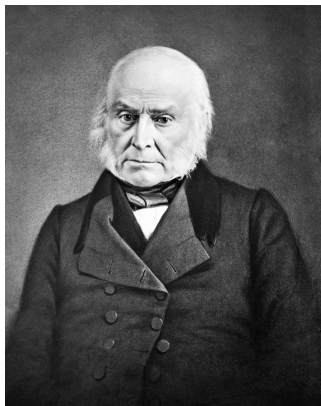
The rule is named after the sixth president of the United States, John Quincy Adams, and the electoral formula he proposed.

Definition

Adams Approval Voting is the composite Thiele rule defined by

1. $\omega_1(\ell) = (1, 0, 0, 0, 0, \dots)$ and
2. $\omega_2(\ell) = (1, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots)$.

We extend it from parliamentary elections to committee voting.



ADAMS APPROVAL VOTING: EXPLANATION

Concretely, the rule proceeds as follows:

1. Select all committees W maximising the number of ‘covered’ voters, i.e., that maximise

$$|\{i \in N : A_i \cap W \neq \emptyset\}|$$

2. Among those, select all committees W' that maximise

$$\sum_{i \in N} \sum_{\ell=2}^{|A_i \cap W'|} \frac{1}{\ell - 1}$$

SUPERCUT: MORE RESULTS ON REPRESENTATION

...

2. MANIPULATION



3. DELIBERATION

